COMP 3005

Database Design — Team 127

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Normalization

The process undertaken has the goals of: minimizing the redundancy presented by repeated data and improving the structural integrity of the database design.

1NF: Requires a disallowing of multivalued attributes, composite attributes, and their combinations, which we do not have. All the attributes within the relations' columns are single atomic (indivisible) values, as required. As well, there are not any duplicate rows (uniqueness for rows), and no nested relations either - our tables are flat, which is what 1NF enforces. Overall, there is no need for decomposition of values (the main violation of 1NF).

The following two 'rules' are concerned with functional dependencies:

2NF: Second normal form requires full functional dependency and this is ensured by eliminating partial dependencies. In every relation of the database, the non-prime (prime being primary) attributes depend on the whole primary key, and not just a part of it. This means that these non-prime attributes are fully functionally dependent on the primary keys. In general, the primary keys we have are all single values and not unique composites of values.

3NF: Third normal form is violated if the schema has transitive dependencies - nonprime attributes depending on other non-prime attributes. Within all the relations of the database design, the functional dependencies are strictly between non-prime and primary attributes, and nothing else. As such, there are no use cases present for decomposition into entities with directly dependent data.